

Dynamic Signal-Tilt Portfolio Strategies

1 Problem and strategy family

We observe daily returns and three signals for $N = 20$ securities:

$$r_{i,t}, \quad s_{i,t}^{(1)}, s_{i,t}^{(2)}, s_{i,t}^{(3)}, \quad i = 1, \dots, N.$$

The return $r_{i,t}$ is a *forward return*: the weight held on day t earns $r_{i,t}$. Therefore the day- t portfolio may use information only from days $t-1, t-2, \dots$. Using $s_{i,t}$ to determine $w_{i,t}$ would be look-ahead bias.

All proposed strategies are variants of one architecture:

equal-weight core + cross-sectional signal tilt + risk-controlled leverage + execution control.

The variants change the signal mixture, smoothing speeds, tilt size, leverage rule, timing overlay, or concentration limits. The signal-free strategy is a control rather than a serious alpha candidate.

2 Notation

For a cross-section $x_t = (x_{1,t}, \dots, x_{N,t})$, define

$$\mathcal{Z}(x_t)_i = \frac{x_{i,t} - \bar{x}_t}{\sqrt{N^{-1} \sum_{k=1}^N (x_{k,t} - \bar{x}_t)^2}}, \quad \bar{x}_t = \frac{1}{N} \sum_{k=1}^N x_{k,t}.$$

Thus $\mathcal{Z}(x_t)$ has cross-sectional mean zero and variance one. It describes relative rankings, not the absolute level of a signal.

For a vector x , define unit-gross normalisation by

$$\mathcal{U}(x) = \frac{x}{\sum_i |x_i|}.$$

A vector satisfying $\sum_i |w_i| = 1$ has gross exposure one.

For half-life h , define the exponentially weighted average

$$\mathcal{E}_h(x)_t = \frac{\sum_{u=0}^t 2^{-(t-u)/h} x_u}{\sum_{u=0}^t 2^{-(t-u)/h}}.$$

An observation's influence is halved after h days. A short half-life reacts quickly but is noisy; a long half-life is smoother but responds slowly.

3 Step 1: causal signal construction

The lagged, standardised signals are

$$z_t^{(j)} = \mathcal{Z}\left(s_{t-1}^{(j)}\right), \quad j = 1, 2, 3.$$

Each signal is smoothed at its own half-life:

$$\tilde{z}_t^{(j)} = \mathcal{E}_{h_j}\left(z_t^{(j)}\right)_t.$$

Given mixture weights $\beta_1, \beta_2, \beta_3$, with $\beta_1 + \beta_2 + \beta_3 = 1$, define the combined score

$$g_t = \mathcal{Z}\left(\beta_1 \tilde{z}_t^{(1)} + \beta_2 \tilde{z}_t^{(2)} + \beta_3 \tilde{z}_t^{(3)}\right).$$

A positive $g_{i,t}$ means that security i ranks favourably relative to the other securities on day t .

Where the half-lives come from

The half-lives are not theoretical constants. They were selected from a coarse, causal grid of nearby smoothing speeds and compared across time splits, ten contiguous folds, transaction-cost simulations, and long block-bootstrap paths.

The evidence supports a region rather than one exact point:

$$\text{signal 1: fast,} \quad \text{signal 2: medium,} \quad \text{signal 3: very fast.}$$

The challenger uses $(h_1, h_2, h_3) = (5, 8, 3)$. These integers should not be interpreted as precisely estimated. Selecting increasingly exact half-lives from the same sample would create overfitting.

4 Step 2: core and signal portfolios

The robust core is equal weight:

$$c_{i,t} = \frac{1}{N}.$$

The unit-gross signal portfolio is

$$a_t = \mathcal{U}(c_t \odot g_t),$$

where $\odot$ denotes elementwise multiplication.

Let $\alpha \in [0, 1]$ be the signal-tilt fraction. The portfolio shape is

$$\phi_t = \mathcal{U}((1 - \alpha)c_t + \alpha a_t).$$

Hence $\alpha = 0$ gives the equal-weight control, while a larger α places more weight on the signal ranking. Normalisation keeps the shape at unit gross before leverage is applied.

Equal weight is used because it is stable, transparent, and difficult to estimate incorrectly. A principal-component core was tested but rejected: PC1 maximises explained variance, not expected growth, and concentrated the portfolio in the dominant high-volatility factor.

5 Step 3: downside-risk leverage

The unlevered paper return of the shape is

$$q_t = \phi_t^\top r_t.$$

Define the downside-risk observation

$$d_t = 2 \min(q_t, 0)^2.$$

The factor 2 places downside risk on approximately the same scale as ordinary variance under a symmetric return distribution. Only losses increase the estimate directly.

The causal risk estimate used on day t is

$$\hat{\sigma}_t^- = \sqrt{\mathcal{E}_{60}(d)_{t-1}}.$$

For a downside-risk target v^* , the preliminary leverage is

$$\ell_t^{(0)} = \text{clip}\left(\frac{v^*}{\hat{\sigma}_t^-}; 2, L_{\max}\right).$$

Thus leverage falls after realised downside volatility increases and rises when estimated downside risk falls. The lower bound avoids abandoning the book after temporary volatility; the upper bound controls ruin risk.

The main target is

$$v^* = 0.0375,$$

meaning 3.75% daily downside risk. This came from a coarse grid over nearby targets. It is an empirical design choice, not a mathematical constant.

During the first 20 observations, leverage is fixed at

$$\min(5, L_{\max}),$$

because the risk estimate has too little history.

6 Step 4: signal-2 market timing

Signal 2 is also used through its market-wide component

$$m_t = \frac{1}{N} \sum_{i=1}^N s_{i,t}^{(2)}.$$

Using only information available by day $t - 1$, m_{t-1} is standardised against 500-day exponentially weighted moments, clipped to $[-3, 3]$, and smoothed with a five-day half-life. Denote the resulting feature by τ_t .

Final leverage is

$$\ell_t = \text{clip}\left(\ell_t^{(0)} + \gamma\tau_t; 2, L_{\max}\right),$$

where γ is the timing amplitude.

This is distinct from using signal 2 to rank securities. The cross-sectional component chooses *which* securities to own; the common component changes *how much* total risk is taken.

7 Why retain signal 2?

Signal 2 does not have strong standalone one-day statistical significance. Its justification is weaker and more cautious:

1. Its predictive relationship is more visible over multi-week horizons than over one day, suggesting slow information.
2. It ranks securities differently from signals 1 and 3, so it can reduce losses when those signals fail together.
3. Its common component contains more convincing market-timing information.

Its portfolio contribution is regime-dependent and differs across sample halves. Therefore its exact weight is not statistically established. The challenger reduces its cross-sectional weight from 40% to 30%. This is a shrinkage and diversification decision, not a claim that signal 2 is proven alpha.

A timing-only signal-2 variant was tested internally. It removed signal 2 from security selection but retained τ_t . It materially reduced CAGR and worsened drawdown and worst-fold performance, so it was not retained as a dashboard candidate.

8 Step 5: position limits

The ideal leveraged weights are

$$w_{i,t}^{\text{ideal}} = \text{clip}(\ell_t \phi_{i,t}; -p_{\max}, p_{\max}).$$

The position cap $p_{\max}$ prevents one security from dominating the book. A separate submitted gross limit is imposed:

$$\sum_i |w_{i,t}^{\text{target}}| \leq 9.5.$$

Short selling and leverage are allowed, but these constraints reduce concentration and keep the strategy below the permitted gross exposure.

9 Step 6: drift, trading and costs

Weights drift after returns are realised. If $w_{i,t-1}^{\text{actual}}$ is yesterday's actual weight and $R_{t-1}^{p,\text{net}}$ is yesterday's net portfolio return, then the weight carried into day t is

$$\tilde{w}_{i,t} = w_{i,t-1}^{\text{actual}} \frac{1 + r_{i,t-1}}{1 + R_{t-1}^{p,\text{net}}}.$$

If the denominator is non-positive, the book is treated as wiped out and the drifted weights are reset to zero.

Let

$$\Delta_{i,t}^{\text{desired}} = w_{i,t}^{\text{ideal}} - \tilde{w}_{i,t}.$$

The causal 20-day volatility estimate of security i is

$$\hat{\sigma}_{i,t} = \sqrt{\mathcal{E}_{20}(r_i^2)_{t-1}}.$$

A volatility-scaled deadband removes small trades:

$$\Delta_{i,t}^{\text{band}} = \text{sign}(\Delta_{i,t}^{\text{desired}}) (|\Delta_{i,t}^{\text{desired}}| - \hat{\sigma}_{i,t})_+.$$

The planned trade is

$$\Delta_{i,t}^{\text{plan}} = \text{clip}(\Delta_{i,t}^{\text{band}}; -0.049, 0.049),$$

and

$$w_{i,t}^{\text{target}} = \tilde{w}_{i,t} + \Delta_{i,t}^{\text{plan}}.$$

The value 0.049 is kept just below the evaluator’s 0.05 trade cap to avoid repeatedly requesting capped trades and paying the cap penalty.

The evaluator applies

$$\Delta_{i,t}^{\text{applied}} = \text{clip}(w_{i,t}^{\text{target}} - \tilde{w}_{i,t}; -0.05, 0.05),$$

so the actual weight is

$$w_{i,t}^{\text{actual}} = \tilde{w}_{i,t} + \Delta_{i,t}^{\text{applied}}.$$

Daily transaction cost is

$$C_t = 0.0005 \sum_i |\Delta_{i,t}^{\text{applied}}| + 0.0001 \sum_i \mathbf{1} \{|w_{i,t}^{\text{target}} - \tilde{w}_{i,t}| \geq 0.05\}.$$

The net portfolio return is

$$R_t^{p,\text{net}} = \sum_i w_{i,t}^{\text{actual}} r_{i,t} - C_t.$$

The first evaluated day is free: its submitted weights are established directly with no transaction cost or trade cap. If $R_t^{p,\text{net}} \leq -1$, the portfolio is bust and later days are discarded.

10 Strategy variants

Strategy	$(\beta_1, \beta_2, \beta_3)$	(h_1, h_2, h_3)	α	v^*	$L_{\max}$	γ	$p_{\max}$	Leverage rule
linear_prod	0.30, 0.40, 0.30	5, 15, 5	0.475	0.0375	6.0	0.50	1.50	Downside risk
linear_lower_s2_fast	0.30, 0.30, 0.40	5, 8, 3	0.475	0.0375	6.0	0.50	1.50	Downside risk
linear_no_timing	0.30, 0.40, 0.30	5, 15, 5	0.475	0.0375	6.0	0	1.50	Downside risk
linear_lower_tilt	0.30, 0.40, 0.30	5, 15, 5	0.400	0.0375	6.0	0.50	1.50	Downside risk
linear_conservative	0.30, 0.40, 0.30	5, 15, 5	0.475	0.0350	5.5	0.25	1.25	Downside risk
linear_fixed_5_5	0.30, 0.40, 0.30	5, 15, 5	0.475	—	5.5	0	1.50	Fixed 5.5
equal_core_5_5	—	—	0	—	5.5	0	1.50	Fixed 5.5

Interpretation

linear_prod

The incumbent. It gives signal 2 the largest signal weight and uses moderately slow smoothing for that signal.

linear_lower_s2_fast

The leading challenger. It shrinks signal 2, increases signal 3, and uses faster smoothing. It is preferred if the future data-generating process is reasonably stable.

linear_no_timing

An ablation testing whether the common signal-2 timing feature adds value.

linear_lower_tilt

Tests whether less dependence on all signals improves robustness if predictive relationships weaken.

linear_conservative

Reduces the risk target, timing amplitude, leverage cap, and position cap. It sacrifices expected growth for lower concentration and tail risk.

linear_fixed_5_5

Keeps the signal portfolio but removes dynamic downside-risk sizing. It tests whether the adaptive leverage rule is genuinely useful.

equal_core_5_5

Contains no signal information. It is the simplest benchmark and measures how much performance comes from the underlying equal-weight market exposure.

11 Why not continuously use a t-stat?

A rolling information-coefficient t-stat can be calculated causally. However, using

$$\alpha_t = \alpha_{\max} \tanh(T_t/k)$$

to resize or reverse the signal portfolio performed poorly in ordinary regimes.

The reason is that rolling t-statistics are noisy. The strategy tends to cut exposure after temporary signal losses, pays transition costs, and may miss the subsequent recovery. A correctly scaled signed t-stat protected the book when all signals were permanently reversed, but materially reduced normal and high-volatility CAGR.

Therefore t-statistics are retained as diagnostics:

- to assess signal stability;
- to compare the first and second halves of the sample;
- to identify possible structural failure;
- to inform shrinkage of signal weights.

They are not used as the main daily sizing rule. The fixed tilt acts as a strong prior and avoids reacting excessively to noisy short-run evidence.

12 Assumptions and limitations

1. **Causality:** day- t weights use only information available by day $t - 1$.
2. **Some stability:** the future process must preserve at least part of the training relationship between signals and returns.
3. **Specified execution model:** transaction costs, drift, trade caps and penalties follow the evaluator rules above.
4. **No Gaussian assumption:** the trading rule does not require normally distributed returns. The factor 2 in downside risk is only a scale convention motivated by symmetry.
5. **Parameter uncertainty:** signal weights, half-lives and risk targets were chosen using the same finite training sample. They are design choices, not known population parameters.
6. **Stress tests are not new data:** folds and block bootstraps measure sensitivity but do not create independent evidence.

13 Complete daily algorithm

For each day t :

1. Lag and cross-sectionally standardise each signal.
2. Smooth signals using their strategy-specific half-lives.
3. Form the weighted signal score and standardise it again.
4. Construct the equal-weight core and unit-gross signal portfolio.
5. Blend them using the fixed signal tilt α .
6. Estimate downside risk using returns observed through $t - 1$.
7. Compute clipped leverage and add the causal signal-2 timing overlay.
8. Apply the per-security position cap.
9. Drift yesterday's actual portfolio using realised returns and costs.
10. Ignore desired trades inside the volatility-scaled deadband.
11. Limit planned trades to 0.049 per security.
12. Rescale submitted gross exposure to at most 9.5.
13. Output finite target weights for every security.

The strategy is therefore not a collection of unrelated rules. Each component has a separate purpose:

core : robust market exposure, signals : cross-sectional selection, timing : market-wide risk adjustment, downside target : adaptive leverage, caps and deadband : survival and cost control.
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